



Fleet Wise — Full Test Plan

Everything to check before real users arrive

20 July 2026 · <https://fleet-wise-delta.vercel.app> · [Print this and tick as you go](#)

Part 0 — Configuration you must finish first

Several tests below cannot pass until these are done. Do them in this order.

0.1 Run the pending database migrations

Supabase dashboard → **SQL Editor** → New query → paste each file's contents → **Run**. Expect "Success. No rows returned".

#	File (in your project, folder <code>supabase/migrations/</code>)	What it does	Done
1	<code>0002_ai_metering.sql</code>	Counts AI questions in the database. Without it the copilot always falls back to offline answers.	☐
2	<code>0003_lock_plan_column.sql</code>	Stops users granting themselves the unlimited plan. Security fix — do not skip.	☐
3	<code>0004_reminders.sql</code>	Adds the reminder opt-out flag and the send log.	☐

0.2 Environment variables (Vercel → Settings → Environment Variables)

Use the "paste .env contents in Key input" trick to add them all at once, then **redeploy** (Deployments → ... → Redeploy) so the running app picks them up.

Variable	Where it comes from	Sensitive?
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Supabase → Project Settings → API Keys → service_role (click reveal, copy). It ignores all security rules, so it lives only on the server.	Yes — mark Sensitive
<code>CRON_SECRET</code>	Any long random string. One generated for you: <code>zkk1nTWucAStykiFfVuTwUQa4vVhzGIHOjBy4ZjieZg</code>	Yes
<code>BREVO_API_KEY</code>	brevo.com → free account → Settings → SMTP & API → API Keys → Generate . 300 emails/day, no domain needed.	Yes
<code>EMAIL_FROM</code>	The address users see as sender. Must be verified in Brevo (Senders → Add a sender , confirm by clicking the link Brevo emails you). Your Gmail works.	No
<code>NEXT_PUBLIC_SITE_URL</code>	<code>https://fleet-wise-delta.vercel.app</code> — used for links and the logo inside emails.	No

Careful with the service-role key. Anyone holding it can read, change or delete every user's data, ignoring all protections. Never paste it into a chat, a screenshot, the browser, or any file that gets committed. If it ever

leaks, rotate it immediately in the Supabase dashboard.

0.3 OpenAI billing

platform.openai.com → **Billing** → **Add payment details** → add ~\$5 credit, and set a monthly spend cap while you are there. Without credit every AI request is refused and the app quietly serves offline answers instead.

Part 1 — Accounts and access SECURITY

#	Test	Expected result	Pass
1	Open the site signed out and type <code>/dashboard</code> in the address bar.	Bounced to the sign-in page. No fleet data ever appears.	☐
2	Repeat for <code>/vehicles</code> , <code>/copilot</code> , <code>/pricing</code> .	All bounce to sign-in.	☐
3	Create an account, leaving "Name your garage" empty.	Signup succeeds; garage is auto-named from your email.	☐
4	Sign up again with the <i>same</i> email.	Clear message: an account with this email already exists.	☐
5	Sign in with a deliberately wrong password.	"Wrong email or password" in a red box — never a raw error or <code>{ }</code> .	☐
6	Use a 5-character password at signup.	Rejected with the 8-character rule explained.	☐
7	Sign out, then press the browser Back button.	You do not get back into the app; you land on sign-in.	☐

Part 2 — Vehicles and service records CORE

#	Test	Expected result	Pass
8	Add a vehicle: plate <code>TRK-012</code> , Ford, F-150, 148200 km.	Appears in the list immediately.	☐
9	Reload the page (F5).	Vehicle is still there — proves it saved to the cloud, not the browser.	☐
10	Sign in on your phone with the same account.	The same vehicle appears. This is the real proof data lives in the database.	☐
11	Try to add a second vehicle with the same plate <code>TRK-012</code> .	Rejected — duplicate plates are blocked per garage.	☐
12	Submit the add form with the plate left empty.	Clear validation message; nothing is saved.	☐
13	Enter letters in the mileage field.	Rejected with an explanation.	☐
14	Open the vehicle, log a service: Oil change, \$95, dated 7 months ago .	Record appears in the history table.	☐

15	Edit that record's cost, then reload.	The new cost persists.	☐
16	Delete the record.	Removed from the list and after reload.	☐
17	Delete a vehicle that has service records.	Vehicle and all its records disappear together — no orphans left behind.	☐

Part 3 — Dashboard intelligence CORE

Set this up first: one vehicle with an oil change **7 months ago** (interval is 6 months, so it is overdue), and one with an oil change **5 months ago** (due within 30 days).

#	Test	Expected result	Pass
18	Open the dashboard.	"Overdue services" shows 1; "Due in next 30 days" shows 1.	☐
19	Check the vehicle counter and this month's cost tile.	Both match what you entered.	☐
20	Add three brake repairs to one vehicle, all within the last 6 months.	An "abnormal pattern" warning names that vehicle and suggests finding the root cause.	☐
21	Give one vehicle far higher costs than the rest.	It is flagged as costing more than double the fleet average.	☐
22	Look at both charts.	Monthly cost bars and top-vehicles chart reflect your data; nothing overflows the screen.	☐

Part 4 — The AI copilot CORE

Read the label under each answer. "AI answer" = the real language model replied. "Offline answer (rules)" = the fallback engine answered, which means OpenAI is unreachable (no credit, no key, or migration 0002 not run). Both are correct behaviour, but only the first proves the AI works.

#	Test	Expected result	Pass
23	Ask: "Which vehicles need servicing this month?"	Names your actual overdue vehicles. Labelled AI answer .	☐
24	Ask: "Which vehicle costs me the most and why?"	Correct vehicle with real figures from your records.	☐
25	Honesty test. Ask about a plate you never entered: "Tell me about ZZZ-999".	Says it has no data for that vehicle. If it invents a history, that is a serious failure — tell me.	☐
26	Ask something unrelated: "What is the capital of France?"	Politely steers back to fleet matters.	☐
27	Ask a follow-up in your own words, e.g. "and the cheapest one?"	Sensible answer; no crash.	☐
28	Ask a question with an empty fleet (new account).	Says there are no vehicles yet and suggests adding one.	☐
29	Send a very long question (paste a paragraph).	Either answers or refuses cleanly — no blank screen.	☐

Part 5 — Plan limits and money REVENUE

#	Test	Expected result	Pass
30	On a Free account, ask the copilot 10 questions , watching the counter in the header pill.	Counts down 10 → 0 accurately.	☐
31	Ask an 11th question.	Input locks; upgrade banner appears. No OpenAI call is made — this is what protects your bill.	☐
32	Reload the page and try again.	Still blocked — the limit lives in the database, not the browser.	☐
33	Add 5 vehicles on a Free account, then try a 6th.	Blocked with an upgrade prompt.	☐
34	Go to Pricing and "Subscribe" to Premium.	Plan badge changes; AI allowance jumps to 200; vehicle limit rises to 50.	☐
35	Ask a question on the new plan.	Works again, counter reflects the bigger allowance.	☐
36	Downgrade back to Free.	Limits return to 5 vehicles / 10 questions.	☐

Remember: "Subscribe" takes no money — it is a simulation until a payment provider is connected. Do not advertise paid plans publicly yet.

Part 6 — Security you should verify yourself SECURITY

These are the tests that protect your data and your wallet. Run them in a terminal (Git Bash). Replace `ANON` with your anon key and `TOKEN` with a signed-in user's token (browser DevTools → Application → Local Storage → the `sb-...-auth-token` entry → `access_token`).

6.1 Nobody can call the AI without signing in

```
curl -X POST https://fleet-wise-delta.vercel.app/api/copilot \
-H "Content-Type: application/json" -d '{"question":"hi"}'
```

Expect: `{"error":"Not signed in"}` — HTTP 401. Anything else means strangers can spend your OpenAI credit.

6.2 Nobody can promote themselves to the unlimited plan

```
curl -X PATCH "https://YOUR-PROJECT.supabase.co/rest/v1/organizations?id=eq.YOUR_ORG_ID" \
-H "apikey: ANON" -H "Authorization: Bearer TOKEN" \
-H "Content-Type: application/json" -d '{"plan":"business"}'
```

Expect after migration 0003: permission denied (HTTP 401/403). **Before it:** HTTP 200 and the plan changes — which is exactly the hole 0003 closes. This is the single most important test on this page.

6.3 One garage cannot see another's vehicles

Create a second account, add a vehicle to each, then in account B run:

```
curl "https://YOUR-PROJECT.supabase.co/rest/v1/vehicles?select=registration" \
-H "apikey: ANON" -H "Authorization: Bearer TOKEN_B"
```

Expect: only account B's own vehicles. Account A's plates must never appear.

6.4 The cron endpoint rejects outsiders

```
curl -i https://fleet-wise-delta.vercel.app/api/cron/reminders
```

Expect: HTTP 401 Unauthorized. Without the secret nobody can trigger mass emails.

Part 7 — Maintenance reminder emails

#	Test	Expected result	Pass
37	Make sure at least one service is overdue, then trigger the job by hand: <code>curl -H "Authorization: Bearer YOUR_CRON_SECRET" https://fleet-wise-delta.vercel.app/api/cron/reminders</code>	JSON summary showing <code>"sent": 1</code> .	☐
38	Check your inbox (and spam folder).	Branded email from "Fleet Wise" listing the right vehicles with correct overdue/due	☐

		badges.	
39	Click "Open my dashboard" in the email.	Lands on your live dashboard, not localhost.	☐
40	Run the same command again immediately.	"skippedAlreadySent": 1 and no second email . This is the anti-spam protection.	☐
41	Untick "Email me maintenance reminders" on the dashboard, then run it again (use a fresh day or clear that garage's reminder_log row).	That garage is skipped entirely.	☐
42	Run it with a garage that has nothing due.	"skippedNothingDue" increments; no pointless email.	☐
43	Wait for the scheduled run (07:00 UTC) or check Vercel → your project → Cron Jobs.	The job appears and shows a successful execution.	☐

Part 8 — Phone, tablet and browser

#	Test	Expected result	Pass
44	On a phone, open every page: dashboard, vehicles, a vehicle's detail, copilot, pricing.	Nothing overflows sideways; text is readable without zooming.	☐
45	Tap the hamburger menu.	Full-height drawer slides in; background dims; tapping outside closes it.	☐
46	Tap into a form field on an iPhone.	The page does not zoom in awkwardly.	☐
47	Use the copilot on the phone.	Chat fills the screen; the input stays reachable above the keyboard.	☐
48	Rotate the phone to landscape; try a tablet if you have one.	Layout adapts; nothing is cut off.	☐
49	Switch your device to dark mode.	The app follows; all text stays readable.	☐
50	Try a second browser (Chrome and Firefox, or Safari).	Same behaviour throughout.	☐

Part 9 — When something fails

Symptom	Most likely cause	Fix
Every answer says "Offline answer (rules)"	No OpenAI credit, or migration 0002 not run	Add billing credit; run 0002. Check Vercel → Logs for the exact error.
AI counter never moves	Migration 0002 not applied	Run it in the SQL Editor.
Signup fails with a 500	Supabase's built-in email limit	Connect Brevo SMTP, or keep "Confirm email" off while testing.
Reminder job returns 503	Missing service-role key or email key	Add the env vars, then redeploy.

Emails land in spam	Sending from a generic address without a domain	Expected on the free setup; buy a domain and verify it in Brevo to fix properly.
Sign-in suddenly fails for everyone	Supabase free project paused after a week idle	Open the Supabase dashboard to wake it; upgrade before launch.

Part 10 — Before you let real users in

1. Turn **"Confirm email" back on** in Supabase (Authentication → Sign In / Providers) once Brevo SMTP is connected — otherwise anyone can register with someone else's address.
2. Delete the leftover `probe...@example.com` test users, then run `delete from organizations where id not in (select org_id from memberships);` to clear orphaned test garages.
3. Rotate the OpenAI key (it was shared in plain text during development) and set a monthly spend cap.
4. Upgrade Supabase and Vercel off the free tiers, so the project cannot pause and you are licensed for commercial use.
5. Connect a real payment provider before advertising paid plans, then delete `set_plan_simulated()` from the database.
6. Add a privacy policy and terms — you are storing other people's email addresses and vehicle data.

Fleet Wise test plan — 20 July 2026. Report anything that fails, especially test 25 (invented data) and test 6.2 (plan escalation).